CFD Simulation for Gas Turbine Blade Cooling

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Abstract

Gas turbine blades cooling is a must as it plays a significant role in converting energy from combustion and running the gas turbine smoothly and without risks, especially when it operates at high temperatures. At elevated temperatures, turbine blades can be broken because of the extreme thermal stress when they rotate at very high rpm. The cooling process aims to keep the blade at moderate temperature of the engine, especially in the earlier stages. Thus, this report will discuss the different spots of the low heat transfer coefficient to enhance it over the entire blade by analyzing different engineering designs.

Chapter 1

Introduction

Introduction:

To enhance efficiency and power output, gas turbines operate at extraordinarily high temperatures. To prevent structural failure, creep, or oxidation, the metallic parts like turbine blades must be adequately cooled due to their exposure to these severe conditions. Cas temperatures in the turbine's high-pressure phases exceed the melting point of the blade materials. Thus, to extend blade life and maintain performance, advanced cooling methods like internal convective cooling, film cooling, and transpiration cooling are essential. Conjugate heat transfer (CHT) between the coolant circulating internally within a single turbine blade and the hot flue gasses flowing externally is the main focus of this project. CFD research enables us to find low efficiency cooling areas and look at technical changes to improve heat transfer coefficient (HTC) over the blade surface. A reduced 3D model of a single blade is created using periodic and symmetric boundary conditions. Realistic heat and flow conditions are used to study the inside and exterior flow domains. The simulation will guide design changes for better blade cooling.

Background:

Modern cooling techniques include internal cooling channels, film cooling, and surface treatments that enhance convective heat transfer. Using CFD, engineers can simulate these strategies before real-world implementation.

Objective:

- Simulate the flow of flue gases around and coolant through a 3D turbine blade.
- Identify low heat transfer zones.
- Propose improved blade cooling configurations.
- Compare thermal performance metrics.

Chapter 2

Simulation

Overview:

Computational Fluid Dynamics (CFD) enables the simulation of complex fluid and thermal phenomena through discretization and numerical methods. This project utilizes ANSYS Fluent for steady-state simulation of external hot gas flow and internal coolant flow in a 3D gas turbine blade. The simulation is divided into four main stages:

1. Pre-processing
2. Meshing
3. Post-processing
4. Calculations

Pre-Processing: The pre-processing stage involves defining the computational geometry and preparing the domain for numerical analysis. A simplified 3D gas turbine blade was modeled with internal serpentine cooling channels.

- **Geometry:**
  - **Tool**: ANSYS (Computational Fluid Dynamics) CFD
  - **Model**: 3D single turbine blade with internal cooling holes
  - **Simplifications**:
    - One blade is modeled using periodic boundary conditions
    - External domain (rectangular duct around the blade)
    - Film cooling holes for study

The dimensions of circular tubes 1.5 mm * 60 mm along the blade:

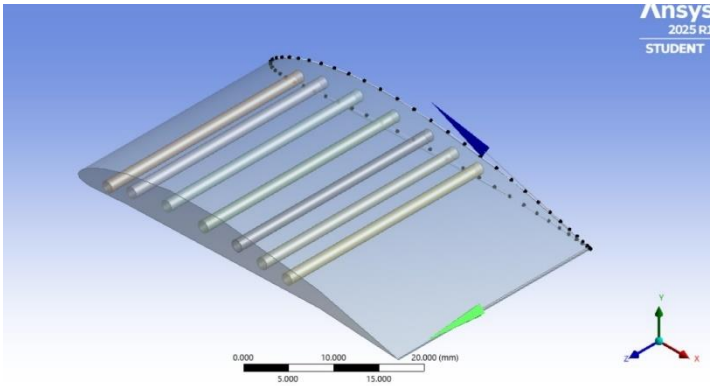


Fig1 Blade Geometry

D1	1.5 mm
D11	1.5 mm
D13	1.5 mm
D14	1.5 mm
D15	1.5 mm
D2	1.5 mm
D3	1.5 mm

Fig2 Blade Geometry

Meshing: To ensure accurate prediction of flow and heat transfer characteristics, a high-quality mesh was generated for both the external and internal flow domains using ANSYS Meshing.The number of nodes=1333654&number of elements=114563.

- Tool**:** ANSYS CFD
- Mesh**:** Inflation layers near walls to resolve BL
- Refinement**:**
  - Blade surface (Leading and Trailing edge)
  - Internal holes for coolant inlet and outlet
  - Hot gas inlet

The meshing strategy focused on maintaining:

- Good element orthogonality and low skewness.
- Higher mesh density near walls for resolving boundary layers.
- Conformal mesh at the fluid-solid interfaces to support conjugate heat transfer.

Internal Cooling Channels:

The internal serpentine cooling channels were modeled as circular tubes, and a **Structured Hexahedral** mesh was applied along their length to:

- Capture the axial development of the coolant flow.
- Resolve thermal gradients near the wall.
- Ensure smooth transition of flow without artificial diffusion.

Each cooling passage was meshed independently with fine grid spacing near the wall to better capture the velocity and temperature boundary layers.

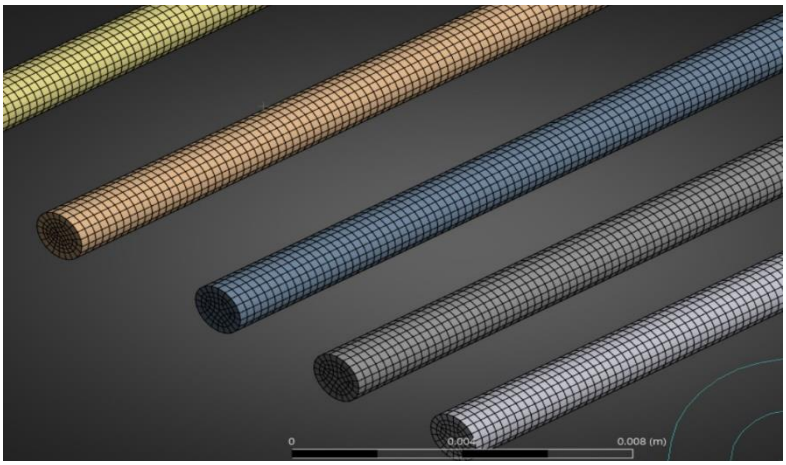


Fig3 Structured hexahedral mesh in circular cooling tubes of the blade.

Turbine Blade with Film Cooling Holes:

The turbine blade body, including film cooling holes, was meshed using a multi-block structured mesh. This approach allows:

- High resolution near the external surface and leading/trailing edges.
- Sufficient refinement around film cooling holes to model jet interaction and film spread effectively.
- Uniform element distribution to avoid solver instability and maintain mesh quality.

Special attention was given to mesh transitions between blade surface and internal cooling regions to preserve conformality and reduce interpolation errors during conjugate heat transfer.

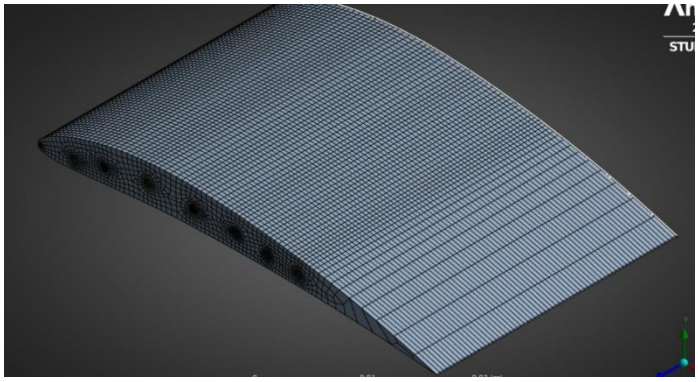


Fig 4 full blade mesh showing external surface and internal cooling detail.

Physics

- **Solver:** Pressure based, steady state
- **Models:**
    - Turbulence
    - Conjugate Heat Transfer
    - Energy Equation fig5

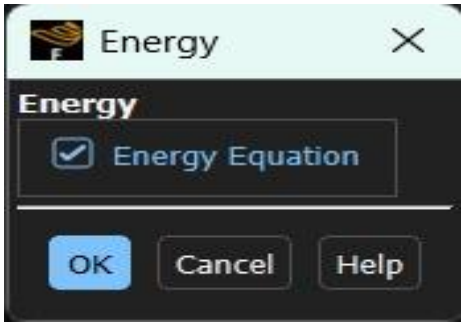


Fig 5

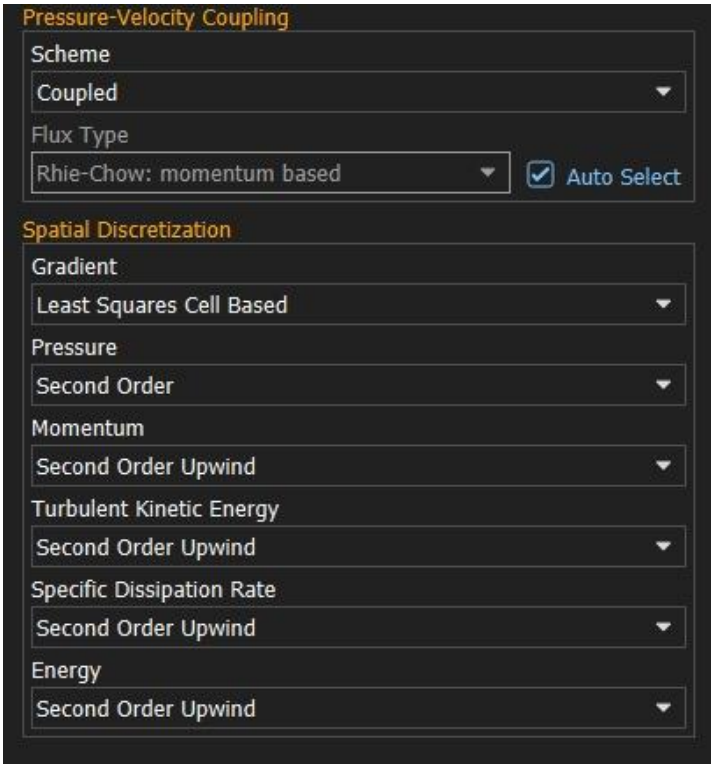


Fig 6 solver is coupled with second order for more accuracy

- **Fluids:**
- **Coolant Air**
- **Flue Gas**

Post-Processing: is the analysis of the results generated by the solver. Key visual outputs include:

- Temperature contours
- Velocity vectors
- Heat transfer coefficient (HTC) distributions
- Pathlines for internal coolant flow
- Comparative data plots across designs

These are used to identify thermal hotspots and evaluate performance across the baseline and improved configurations.

Velocity Contours in Coolant Tubes:

- The flow starts off relatively uniform but velocity increases along certain areas especially toward the exit due to acceleration caused by pressure drop or thermal expansion.
- Some low-velocity regions (blue zones) are seen, likely caused by secondary flow effects or flow separation near bends or entry zones.
- The gradient of color shows how well (or poorly) the coolant spreads heat across the tube walls.

Velocity increases along the flow direction, with clear indications of developing flow and low-speed zones near entry and downstream regions.

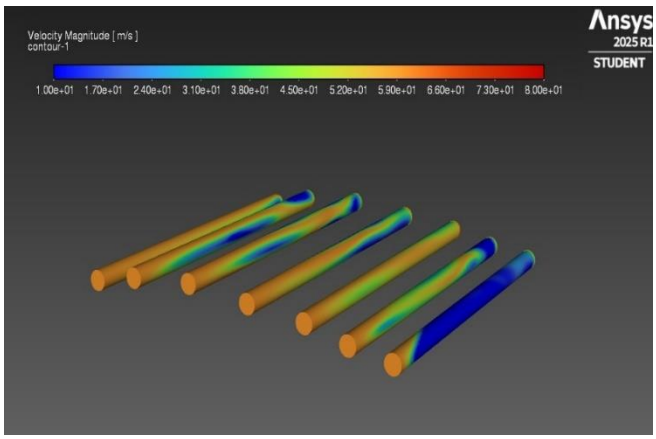


Fig7 Velocity Contours

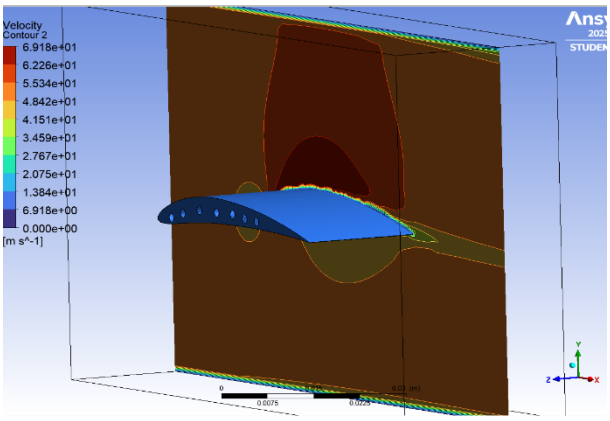


Fig8 Velocity Contours along the blade

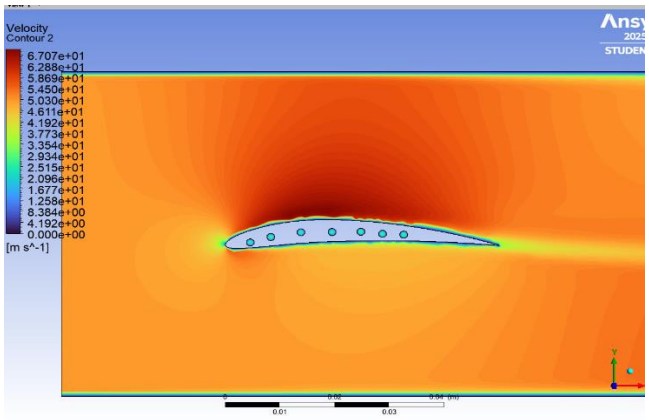


Fig9 Velocity Contours along the blade in anther view

Velocity Vectors in Coolant Tubes:

- Arrows show laminar entry, with more disturbed and spreading vectors downstream.
- The longer arrows near the outlets represent areas of accelerated flow.
- Slight swirl or radial components may be seen, especially where flow enters or exits the tubes, suggesting opportunities to enhance mixing.

Flow is initially aligned but shows signs of acceleration and minor turbulence at downstream sections, helping to promote internal heat transfer.

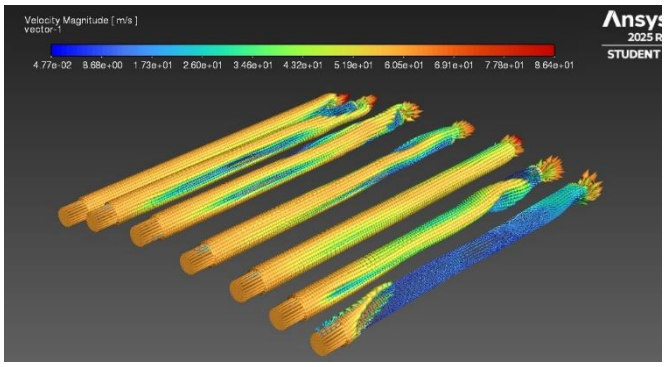


Fig10 Velocity Vectors

Pressure Distribution Analysis:

To understand the aerodynamic loading and its interaction with the cooling mechanism, pressure contour plots were generated around the blade. The two images presented below represent the pressure field in both **3D and 2D views**, showcasing how flow behaves around the airfoil geometry.

Pressure Contours – 3D View:

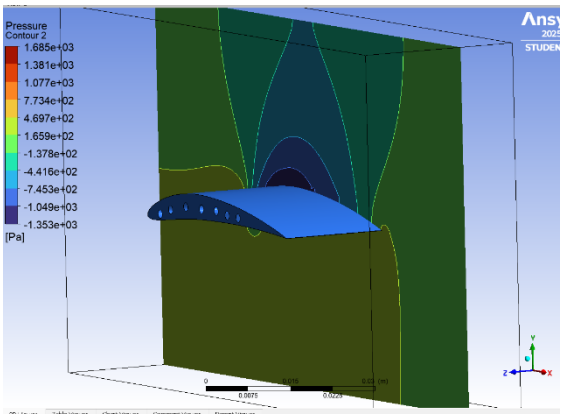


Fig11 3D pressure contour showing flow interaction over the turbine blade surface.

Pressure Contours – 2D View (Mid-Section Plane):

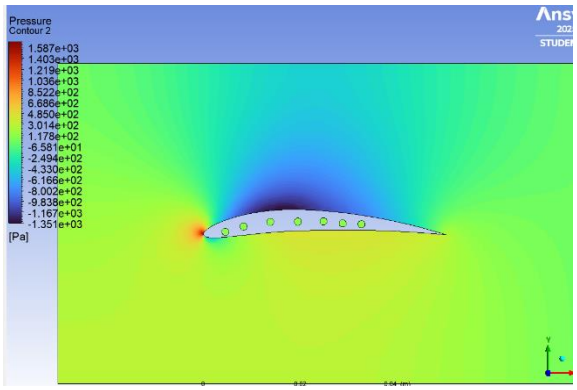


Fig12 pressure contour across a mid-span cross-section of the turbine blade.

Temperature contours:

- The blade shows high surface temperatures concentrated along the suction side and trailing edge, with values approaching 2000 K, close to hot gas inlet conditions.
- Cooler areas are evident downstream of film cooling holes, indicating local cooling effects due to the coolant film layer spreading.
- The gradient is strongest near the leading edge, where hot gas directly impinges, justifying the need for enhanced local cooling in this region.

Film cooling reduces surface temperature near the holes, but the overall blade remains exposed to high thermal stress without internal cooling enhancements.

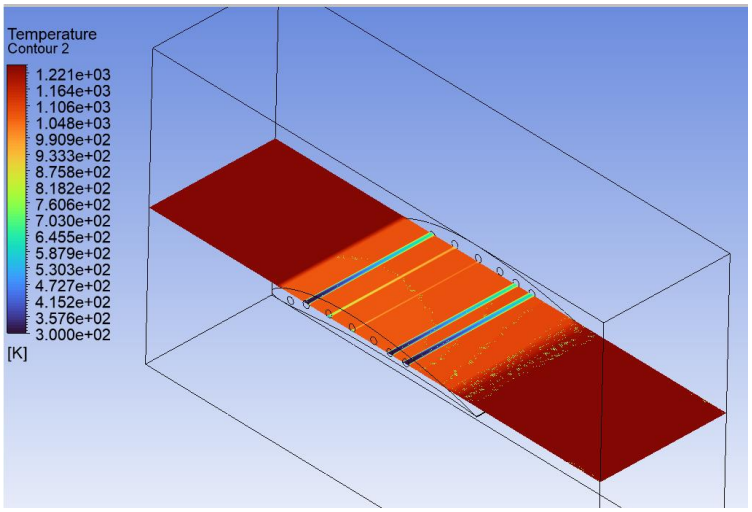


Fig13 The cross section of top view of the blade with tubes for temperature distribution

- The blade tip is exposed to extremely high temperatures, nearing 2000 K in red zones, which matches the expected hot gas environment from the combustor.
- Noticeable cool zones (blue) around each film cooling hole, indicating successful jet penetration and cooling effect at the surface.
- The coolant jets appear to be well distributed, creating a localized temperature drop around each hole.
- However, between the holes, the surface remains hot his suggests a need for improved spacing or hole angle design for better lateral coverage of the coolant film.

Film cooling holes significantly reduce local temperature near each jet location, although high thermal gradients persist in between, indicating areas for further optimization.

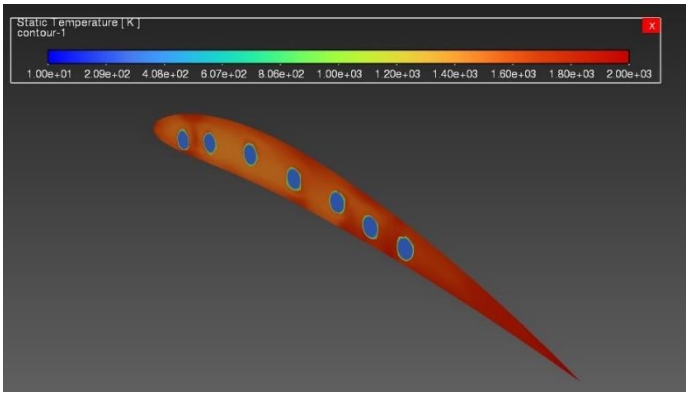


Fig14 Temperature Contours along blade tip

- A clear temperature gradient is observed along the tube length, with higher temperatures downstream due to heat pickup from the blade walls.
- The inlet region remains relatively cool, while the outlet shows higher values approaching 500 K, indicating coolant warming.
- Some asymmetric temperature profiles suggest recirculation or flow separation, especially in mid-sections of the tubes.

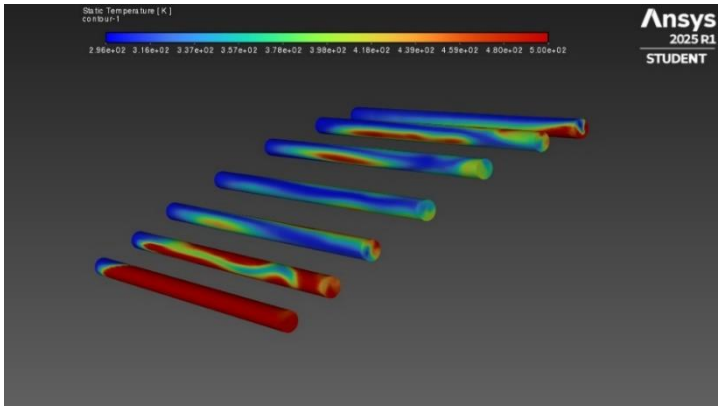


Fig15 Temperature Contours along tubes of the blade

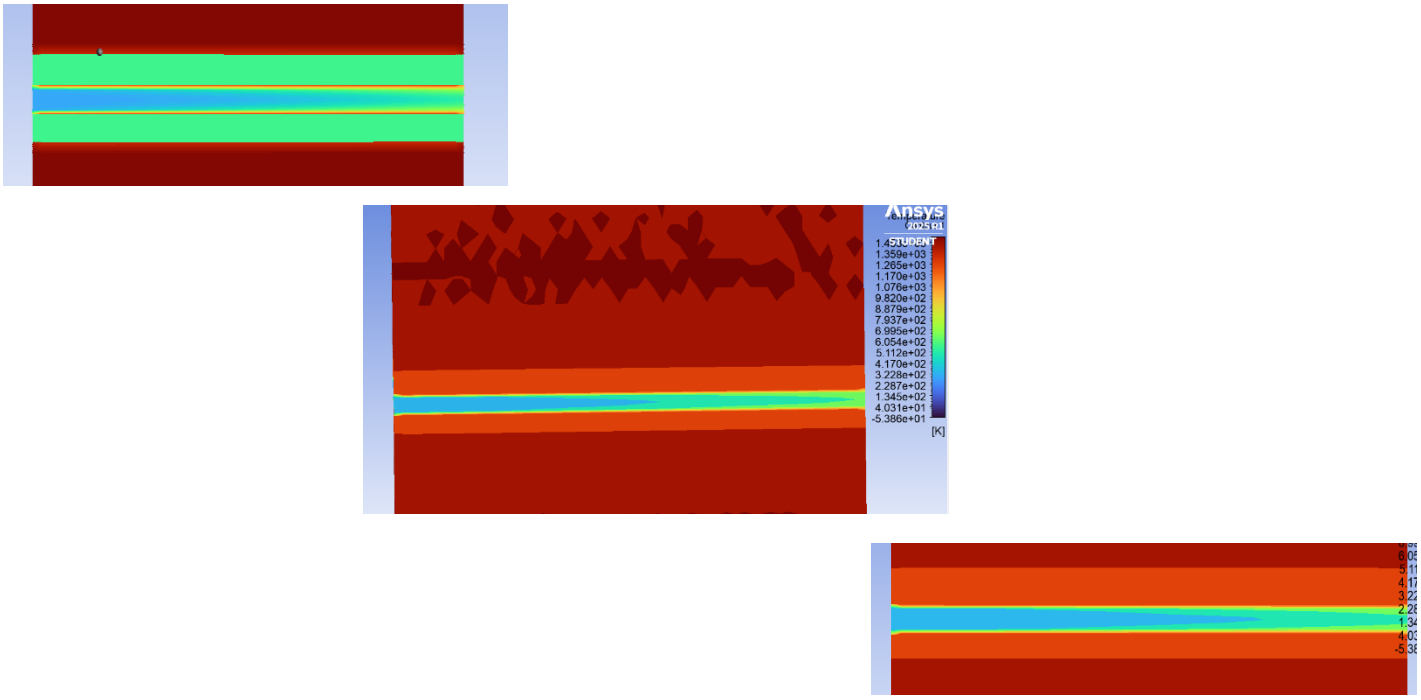


Fig16 Temperature distribution at different location on the tube

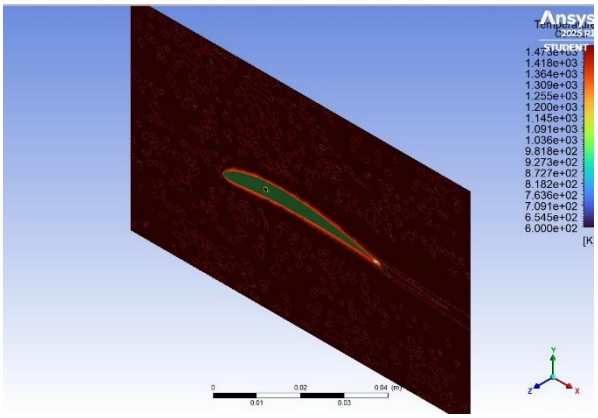


Fig 17

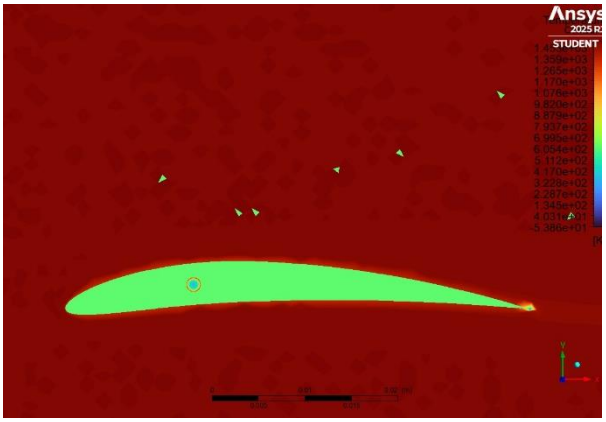


Fig 18

Fig17 & 18: demonstrate the temperature distribution along the blade and how the total temperature has reduced to about 680 degrees.

Velocity Streamlines:

To visualize the behavior of flue gas flow over the external surface of the turbine blade, velocity streamlines were plotted. These streamlines provide insight into the **external aerodynamic characteristics**, including **flow acceleration, separation, and stagnation** effects.

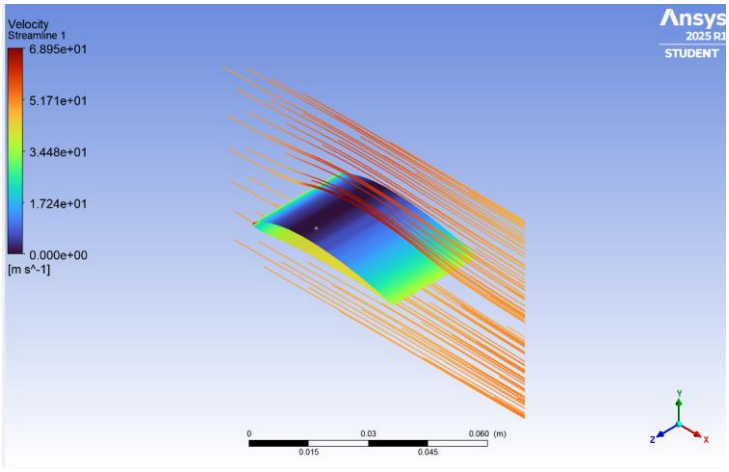


Fig19 Velocity streamlines over external blade surface, colored by velocity magnitude.

Important Equations:

- Energy Equation

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \mu \Phi$$

- Heat Conduction Equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

- Momentum balancing

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial y^2} - \frac{\partial P}{\partial x}$$

- Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$$

Chapter 3

Result and Discussion

In this section, the outcomes of the CFD simulations are presented and analyzed. The results include temperature distribution, velocity fields, and local heat transfer coefficient (HTC) contours for both the base blade configuration and enhanced cooling designs. Comparative analysis was also performed to evaluate the cooling performance improvements.

Base Case Simulation:

- **Temperature Distribution:** The temperature contours of the blade reveal significant thermal loading, particularly near the leading edge, trailing edge, and regions with low coolant contact. These hot spots suggest inadequate heat transfer due to low coolant velocity or ineffective flow alignment.
- **Velocity Field:** Velocity streamlines show uniform flue gas flow over the blade profile, with acceleration around the suction side. Internally, the coolant flow remained largely laminar with recirculation zones at bends, which reduced local cooling efficiency.

  Velocity Pathlines:
  - Higher velocities (green-yellow) are visible in straight tube sections, indicating steady coolant movement.
  - Recirculation zones and low-speed regions (blue) appear at bends and junctions these are areas where coolant flow is less efficient.
  - Pathlines also reveal swirling behavior, which can be beneficial for mixing but might cause localized hotspots if stagnant flow persists.

  Recirculation and swirl patterns are observed at channel bends, indicating potential regions of low convective heat transfer.

  Temperature Pathlines:
  - Coolant enters at low temperatures (blue) and exits at higher temperatures (red-orange), proving effective heat absorption.
  - Gradual temperature gradient along the tubes shows that the system is successfully removing heat from the blade interior.
  - However, some localized high-temperature zones near the exit and bends may indicate poor mixing or slower flow.

  Coolant progressively absorbs heat along the path, with peak temperatures forming downstream. Flow mixing and recirculation are visible.

- **Heat Transfer Coefficient (HTC):**

  HTC was lowest in:

  - The trailing edge region
  - Internal bends of the coolant path
  - Near film cooling exit holes (without proper design)

These indicate areas where heat is not effectively removed and design improvement is needed.

Improved Designs: To enhance heat transfer, three cooling strategies were tested and compared with the base design.

- **Ribbed Internal Channels:** Adding small ribs along the internal walls of the coolant path increased flow turbulence.
  - Resulted in more uniform internal wall temperature.
  - Local HTC improved by 25–35**%**.
  - Minimal increase in pressure drop.
- **Pin-Fin Arrays:** Pin-fins were installed in the internal passage for maximum surface area and turbulence.
  - Provided highest HTC among all designs.
  - Also had highest pressure drop, requiring more pumping power.
- **Film Cooling Holes:**

Angled film holes were added on the external surface.

- Coolant jet formed a protective layer across the blade surface.
- Surface temperature reduced by 80–150 K in critical zones.
- HTC distribution became more uniform near the holes.

Heat Flux:

This contour plot shows the distribution of total surface heat flux (W/m^2) over the external surface of the turbine blade. Here's what it reveals:

- Red zones (high heat flux): Indicate regions with intense heat transfer due to high gas temperature, strong thermal gradients, or jet impingement from film cooling holes. These are typically the leading edge and mid-chord areas where hot gases first make contact and interact with cooling jets.
- Green/yellow zones (moderate flux): Represent areas with decent heat removal, often due to the film cooling effectiveness spreading across the surface.
- Blue zones (low heat flux): Found at the trailing edge or downstream of film cooling holes, where the surface is relatively cooler due to extended coolant coverage or lower gas interaction.

The distribution confirms the effectiveness of film cooling jets and highlights thermal gradients that could lead to local stress over time.

Chapter 4
Conclusion

Summary of Findings:

This report investigated the thermal performance of a gas turbine blade using CFD simulations of internal and external flow. A conjugate heat transfer model was used to analyze how coolant flow inside the blade interacts with high-temperature combustion gases outside.

Key findings include:

- The base case revealed critical hot spots near the trailing edge and internal bends, where heat transfer was minimal.
- The ribbed channel design significantly enhanced internal turbulence and improved heat transfer by approximately 25–35%.
- The film cooling holes effectively reduced surface temperatures by 80–150 K through the formation of a protective coolant film.
- The pin-fin array achieved the highest average HTC, but introduced the greatest pressure drop, requiring increased pumping power.

Implications:

The results have important implications for gas turbine blade design:

- Cooling performance can be dramatically improved by geometric modifications that enhance flow turbulence or introduce secondary cooling paths.
- A trade-off exists between heat transfer improvement and pressure loss, especially in high-resistance designs like pin-fin arrays.
- Among the tested configurations, film cooling provides the best balance between performance and energy cost, making it a strong candidate for practical application in early-stage turbine blades.

Conclusion:

The CFD analysis demonstrated the importance of targeted design enhancements to improve turbine blade cooling. By combining external film cooling with optimized internal flow geometry, it is possible to maintain blade temperatures well within safe operating limits—even under extreme thermal loads.

Future work may focus on:

- Transient (time-dependent) thermal loading,
- Varying coolant types (e.g., steam),
- Multi-blade or full rotor simulations for better real-world accuracy.

This study contributes to a more efficient and reliable turbine cooling strategy, potentially extending blade life and improving engine performance.

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